Use a pencil or a pen with black or dark blue ink. Do NOT write your name. Do NOT write outside the box.

Question 2

Part A

i. $2.883\,g\,H_2O \cdot \frac{1\ mol\ H_2O}{18.02\,g\,H_2O} = 0.1600\ mol\ H_2O$

ii. $0.2400\ mol\ CO_2 \cdot \frac{1\ mol\ C}{1\ mol\ CO_2} = 0.2400\ mol\ C$

$0.1600\ mol\ H_2O \cdot \frac{2\ mol\ H}{1\ mol\ H_2O} = 0.3200\ mol\ H$

$\frac{0.3200\ mol\ H}{0.2400\ mol\ C} = \frac{4\ mol\ H}{3\ mol\ C}$

Empirical formula = $\boxed{C_3H_4O_3}$

Part B

i. $16.0\ mL\ NaOH \cdot \frac{1\ L}{1000\ mL} \cdot \frac{0.0550\ mol\ NaOH}{1\ L} = 8.80 \cdot 10^{-4}\ mol\ NaOH$

$\frac{8.80 \cdot 10^{-4}\ mol\ HAsc}{10.0\ mL\ HAsc} \cdot \frac{1000\ mL}{1\ L} = \boxed{0.0880\ M\ of\ HAsc}$

ii. Equivalence point = 16.0 mL → Half-equivalence point = 8.0 mL

At 8.0 mL, pH = pKa → $\boxed{pK_a \doteq 4.1}$

iii. $pH = pK_a + \log_{10}\left(\frac{[Asc^-]}{[HAsc]}\right)$

$4.7 = 4.1 + \log_{10}\left(\frac{[Asc^-]}{[HAsc]}\right)$

$0.6 = \log_{10}\left(\frac{[Asc^-]}{[HAsc]}\right)$

$10^{0.6} = \frac{[Asc^-]}{[HAsc]}$

$\boxed{\frac{[Asc^-]}{[HAsc]} = 4.0}$

Question 2 is continued on the next page.

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Question 2

Part C

i. Using trials 1 and 3, we can see that the reaction rate is 2 times greater when [HAsc] is 2 times greater, so it is first order with respect to [HAsc].

ii. $2.457 \cdot 10^{-4} \frac{M}{s} = k \cdot 0.450M \cdot 1.200M$

$$\boxed{k = 4.55 \cdot 10^{-4} s^{-1} M^{-1}}$$

Part D

Because I_3^- is an ion unlike I_2, it experiences ion-dipole forces with H_2O. The I_3^- ion is attracted to the partially positive H atoms in H_2O, making I_3^- soluble.

Continue to Question 3.